

Team Project Proposal
Cloud-Based IoT Environmental Telemetry Platform
Seid Cubro - Charles Shoppel

Team Members:

- Charles Shoppel
- Seid Cubro

Member Roles:

- Project Lead / Cloud / Infrastructure / Security: Seid Cubro
 - Seid Cubro will be responsible for leading the project as well as developing the cloud aspect, project infrastructure and security features.
- Application / Dev Ops: Charles Shoppel
 - Charles Shoppel will be responsible for building and containerizing the application services and ensuring reliable builds and deployments using Docker, Kubernetes and CI/CD workflows.

Project Description:

This project will implement a cloud-based IoT Environmental Telemetry Platform that ingests temperature and humidity data from IoT sensors, processes the data through containerized microservices and exposes it via secure REST APIs. The system is deployed on a cloud platform using Infrastructure as Code and includes automated CI/CD pipelines, centralized logging, monitoring, alerting, and security controls.

Project Goals:

Design and deploy a cloud-native microservice architecture for IoT data ingestion and retrieval, implement containerized services with independent deployment and scaling, use Infrastructure as Code to provision and tear down cloud resources reproducibly, build a CI/CD pipeline that automates testing, building and deployment, implement centralized logging, metrics and alerting to monitor system health, secure the system using authentication, secrets management, and least-privilege access, demonstrate system reliability through failure scenarios and recovery.

Target Usage:

This project will be intended for everyday use in any environment. The project will track temperature and humidity frequently, giving a detailed look at one's environment in order to assess comfort, habitability, etc. The goal of the project is to add frequent telemetry ingestion

with great security, runtime and reliability. The project will mainly be used in one's home. Target users will be homeowners and everyday people.

Constraints:

Team size limited to two people, limiting the scale of the project. Project must be completed within the course timeline, about three months. Cloud resources must be within budget.

Equipment Requirements:

Cloud account with AWS, GitHub repository for CI/CD, Docker and Docker Compose for local development, Terraform for infrastructure provisioning, two ESP32 boards, two DHT11 sensors.

Development Tools:

- Cloud Service: AWS with ECS Fargate
- Database: DynamoDB
- Infrastructure as Code: Terraform
- CI/CD: GitHub
- Observability: CloudWatch Logs and Alarms
- Security: AWS Secrets Manager, IAM Roles per Service, JWT Authentication